Jakaria Chowdhury Tajwone

Aspiring AI Researcher | Machine Learning · NLP · Computer Vision | Full-Stack Developer
+880-1342-066856 | tajwone248tc@gmail.com | 0562310005101031@neub.edu.bd | Sylhet, Bangladesh
GitHub | LinkedIn | Portfolio | Codeforces

Professional Summary

Computer Science & Engineering student (CGPA 3.81/4.00) focused on machine learning, deep learning, and NLP for low-resource and code-mixed languages. First author of "SylHate", a hate speech detection paper on Romanized Sylheti accepted and presented at IEEE COMPAS 2026. Experienced in Python, Scikit-learn, TensorFlow/Keras, and transformer models, with a strong base in algorithms (550+ competitive programming problems; Codeforces max rating 1183) and full-stack development. Seeking a Student Researcher position to contribute to transparent, trustworthy AI research.

Research Interests

- **Natural Language Processing:** low-resource and code-mixed languages, hate speech detection, text classification, transformer-based models.
- **Machine Learning & Deep Learning:** hybrid neural architectures (BiLSTM/BiGRU, transformer-based), comparative benchmarking, uncertainty-aware inference.
- **Computer Vision:** deep learning for visual recognition and intelligent sensing systems.
- **Also interested in:** Explainable and Trustworthy AI, Generative AI, Cybersecurity.

Education

Bachelor of Science in Computer Science & Engineering 2023 – Present
North East University Bangladesh | CGPA: 3.81/4.00 (after 7th semester) *Sylhet, Bangladesh*
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Technical Writing & Research Methodology, Data Structures & Algorithms, Database Management Systems, Software Engineering

Publications

[C1] J. C. Tajwone (first author), Mst. F. Jaina, S. S. R. Akash, and A. Ahmad, "SylHate: Hate Speech Detection in Romanized Sylheti Using Machine Learning and Transformer Baselines," *3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*, University of Dhaka, 9–10 October 2026. Organized by IEEE Computer Society Bangladesh Chapter. *Status: Accepted and presented.*

- Studied hate speech detection for Romanized Sylheti, a low-resource language variety, benchmarking classical machine learning models against transformer-based baselines.

Technical Skills

Programming Languages: Python, C, C++, JavaScript (ES6+), SQL
ML / DL: Scikit-learn, TensorFlow/Keras, PyTorch (working knowledge), Transformers, NLP, BiLSTM/BiGRU, CNNs
Research: Experimental design, benchmarking, dataset curation, technical writing
Web & Databases: React.js, Next.js, Node.js, Express.js, RESTful APIs, Tailwind CSS, MySQL, MongoDB
Tools: Git, GitHub, Jupyter Notebook, LaTeX, npm
Core Competencies: Data Structures & Algorithms, Problem-Solving, Full-Stack Development, Research Collaboration, Communication

Technical Projects

CourseSphere – University Course Registration System GitHub

- Developed a course registration platform as Team Lead with role-based access control for students, advisors, HODs, and administrators.
- Architected a multi-tier JWT authentication system with automated email notifications across 5 user roles.
- Built a full-stack application with Next.js, TypeScript, and MySQL featuring course browsing, prerequisite verification, payment processing, and real-time registration tracking.
- Implemented administrative workflows including course approval, financial calculations, result management, and deadline control.

SkyTrip – Online Flight Booking System GitHub

- Engineered a full-stack flight booking platform using React.js, Node.js, Express.js, and MySQL with secure JWT authentication.

- Developed RESTful APIs for flight search by destination, class, and date with real-time seat availability and booking confirmation.
- Implemented secure payment gateway integration, automated confirmation, and an admin panel with complete CRUD operations and booking analytics.

Daily-Flow – Task Management Application GitHub

- Built a productivity app using Next.js, MongoDB, and Tailwind CSS with authentication and task categorization.
- Implemented priority-based organization, progress tracking (daily, weekly, monthly views), and search/filter capabilities in a responsive interface.

TextUtils – Text Manipulation Tool GitHub

- Created a React-based text processing application with 10+ utility features using hooks and a component-based architecture with a Bootstrap UI.

Professional Experience

Frontend Developer Intern August 2025 – October 2025
Quadiz *Remote*

- Contributed to frontend development projects using modern web technologies and frameworks.
- Implemented responsive user interfaces and participated in code reviews following clean-code best practices.

Certifications & Professional Training

Web Development Essential – EDGE Digital Skills Training Dec 2024 – Jan 2025
Bangladesh Computer Council (BCC), ICT Division, SUST CSE *Government Initiative*

- Completed an intensive 2-month full-stack program (HTML5, CSS3, JavaScript, Node.js, Express.js, MySQL, Git).
- Developed and deployed a production-ready full-stack web application in a team using agile practices and code reviews.

Problem-Solving & Achievements

- **Competitive Programming:** Solved 550+ algorithmic problems across multiple platforms.
- **Platforms:** Codeforces (Rating: 1183), CodeChef (Rating: 1446), LeetCode (20+ problems).
- **Core Strengths:** Dynamic Programming, Graph Algorithms, Greedy Algorithms, Sorting & Searching.